Compound Measures

Exam code: 0607

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Written by: Naomi C
Reviewed by: Dan Finlay
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Compound Measures

What is a compound measure?

- **A compound measure** is something that is calculated by using **more than one** measurement
- Compound measures can be used to **measure rates**
  - This measures how much one quantity changes when the other is increased by 1
  - Examples include:
    - **Speed** – how much the distance changes for each unit of time
    - **Flow rate** – how much the volume changes for each unit of time
    - **Population density** – how many people there are for each unit of area
    - **Fuel consumption** - volume of fuel used for each unit of distance travelled

How do I find the units for a compound measure?

- You can use the **formula** for a compound measure to **derive its units**
  - Use the **units** for the **quantities** in the formula to derive the units of the compound measure
  - Write a **division** as **a/b** and pronounce it as "a per b"
- Examples include:
  - $\text{Speed} = \dfrac{\text{Distance}}{\text{Time}}$
    - If the distance is measured in **km** and the time is measured in **minutes** then the speed is measured in **km/min**; kilometres per minute
  - $\text{Flow rate} = \dfrac{\text{Volume}}{\text{Time}}$
    - If the volume is measured in **m^3** and the time is measured in **minutes** then the flow rate is measured in **m^3/min**; metres cubed per minute

How do I find the formula for a compound measure?

- You can use the **units** for a compound measure to help remember its **formula**

Help me understand

Stuck on something?

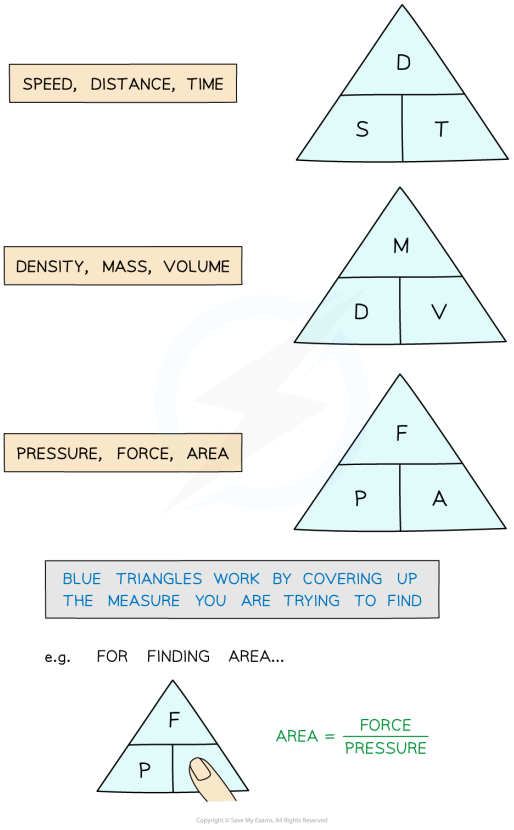
Ask questions and get help with exactly what you're revising.

Got it

measures
- Examples include:
  - **Density** can be measured in **kg/cm^3**
  - kg is a measure of **mass** and cm^3 is a measure of **volume**
  - Therefore $\text{Density} = \dfrac{\text{Mass}}{\text{Volume}}$

What is a formula triangle?

- A **formula triangle** shows the **relationship** between the **different measures** in a **compound formula**
- They can be useful if you are less confident with rearranging formulas
- Three examples are shown below
  - **Speed** is equal to **distance** divided by **time**
  - **Density** is equal to **mass** divided by **volume**
  - **Pressure** is equal to **force** divided by **area**



Examiner Tips and Tricks

- Check in the exam to see if the answer needs to be in different units
  - For example, the question may use metres and seconds but want the answer in km/h
- You need to remember the relationship between **speed**, **distance** and **time**

Worked Example

A high-speed racing car has an average fuel consumption of 3 km per litre during a race.

1 lap of the racing circuit is 5.9 km in length.

(a) Calculate the volume of fuel used, in litres, to complete 15 laps of the circuit.

The units for the fuel consumption are km per litre, which suggests the formula is $\text{fuel consumption} = \dfrac{\text{distance}}{\text{volume}}$

Calculate the total distance covered for the 15 laps

$$15 \times 5.9 \text{ km} = 88.5 \text{ km}$$

Use the above formula to find the volume of fuel, V litres, used

$$\text{fuel consumption} = \frac{\text{distance}}{\text{volume}}$$

$$3 \text{ km per litre} = \frac{88.5 \text{ km}}{V \text{ litres}}$$

Rearrange the equation by multiplying both sides by V, and dividing both sides by 3

$$3V = 88.5$$

$$V = \frac{88.5}{3} = 29.5$$

29.5 litres of fuel

The race car then requires a pit-stop to refuel to complete the final laps of the race.

The flow rate of the fuel pump is 720 litres per minute, and fuel is pumped into the car for 3.1 seconds.

(b) Calculate the volume of fuel, in litres, pumped into the car in this time.

The flow rate is 720 litres per minute which suggests the formula is $\text{flow rate} = \dfrac{\text{volume}}{\text{time}}$

Before we can use the formula, we need to change the units of time to both be the same

Change 720 litres per minute, into litres per second (to match the time fuel is pumped for, which is in seconds)

If 720 litres are pumped in 1 minute, 60 times less will be pumped in 1 second

$$720 \div 60 = 12 \text{ litres per second}$$

Substitute these values into the formula

$$12 \text{ litres per second} = \frac{\text{volume in litres}}{3.1 \text{ seconds}}$$

Multiply both sides by 3.1

$$\text{volume in litres} = 12 \times 3.1 = 37.2$$

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